
GENERAL RELATIVITY

Tutorial problem set 12, 06.12.2019.

■ **PROBLEM 32** Free fall into a black hole.

Consider the Schwarzschild metric,

$$ds^2 = -A(r)dt^2 + \frac{dr^2}{A(r)} + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2), \quad (32.1)$$

where

$$A(r) = 1 - \frac{2MG_N}{r}. \quad (32.2)$$

- (a) Generally, if the metric does not depend on one of the coordinates x^α , then the momentum component, $p_\alpha = \partial\mathcal{L}/\partial\dot{x}^\alpha$ is constant along the geodesic parametrized with the affine parameter σ , $dp_\alpha/d\sigma = 0$. That follows from the Euler-Lagrange equation for the Lagrangian $\mathcal{L} = \frac{1}{2}g_{\mu\nu}\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma}$.

Show that for Schwarzschild metric the following quantities

$$E = A(r)\frac{dt}{d\sigma}, \quad L = r^2 \sin^2\vartheta \frac{d\varphi}{d\sigma} \quad (32.3)$$

are conserved along a geodesic parametrized by σ .

- (b) Because of spherical symmetry, the free fall of a particle in Schwarzschild space-time is always confined to a plane. Choose $\vartheta = \pi/2$ (equatorial plane), and express the Schwarzschild line element as a function of E , L , $A(r)$, $dr/d\sigma$, r^2 , and $d\sigma^2$.
- (c) Show that the quantity

$$\epsilon = -g_{\mu\nu}(x)\frac{dx^\mu}{d\sigma}\frac{dx^\nu}{d\sigma} \quad (32.4)$$

is constant along the geodesic (in general space-times). One has $\epsilon = 0$ for light-like trajectories, and one can choose the affine parameter σ such that $\epsilon = 1$ for a massive particle (in which case σ is the proper time τ), and $\epsilon = -1$ for space-like geodesics.

Express $(dr/d\sigma)^2$ as a function of $A(r)$, E , L , and ϵ .

Consider now an object falling radially into the black hole. Its speed at a distance $r = R$ from the center of the black hole is $dr/d\tau = U$ at $t = 0$.

- (d) Express the constant E on such a trajectory in terms of M , R and U .
- (e) Compute the 4-velocity of the falling observer as a function of the parameters E and M , and the coordinate r .
- (f) How much proper time $\Delta\tau$ does it take for the observer to reach the horizon that is located at $r = 2MG_N$? For simplicity set $E = 1$.
- (g) Argue that an external static observer sees that the object reaches the horizon in an infinite amount of time Δt .

■ **PROBLEM 33** Gravitational redshift in Schwarzschild space-time.

Consider two observers in Schwarzschild space-time, at fixed spatial coordinates $(r_1, \vartheta_0, \varphi_0)$ and $(r_2, \vartheta_0, \varphi_0)$ outside of the Schwarzschild radius R_S . Suppose that observer 1 (let's call him Marko) emits a light signal radially towards observer 2 (let's call him Bruno).

- (a) How are the proper times of these observers related to the coordinate time t ?

Label the proper time interval (as measured by Marko) between the emission of two light crests by $\Delta\tau_1$. What is the relation between $\Delta\tau_1$, and Δt , where Δt is the interval of coordinate time between the emission of two light crests (you can view Δt as the time elapsed between the emission of two crests as measured by an observer at $r \rightarrow \infty$).

- (b) What is the proper time interval $\Delta\tau_2$ (as measured by Bruno) between the two successive crests received? Express your result in terms $\Delta\tau_1$, R_S , r_1 and r_2 .
- (c) Compute the ratio of the two frequencies ν_1 (emitted by Marko), and ν_2 (received by Bruno), and show that in the limit $r_1, r_2 \gg R_S$ one has,

$$\frac{\nu_2}{\nu_1} = 1 + \phi(r_1) - \phi(r_2) , \quad (33.1)$$

where $\phi(r_1)$ and $\phi(r_2)$ are the gravitational potentials at the respective coordinates.